Version 5.3 - Smart University Stadium Ticketing

Accounts activation / student-group reservations / secure QR entry / GitHub-safe releases

RELEASE HIGHLIGHTS

Capability	Version 5.3 result
Administrator access	One-time key (20 minutes) activates an account and sets a reusable password. Later logins need email/password; no admin registration or Notepad editing.
Student and group booking	The purchaser receives the first seat. Additional seats require a registered attendee name plus Student ID. One live ticket per student per event.
Individual QR tickets	Every attendee receives their own signed, database-validated QR pass. QR codes do not embed personal information.
Stadium gate operations	Authorized staff scan tickets, verify university identity, and confirm entry separately; duplicate, cancelled and invalid tickets are rejected.
Polished interface	Campus Light and Stadium Night themes, smooth animation, responsive cards and scanning screens; reduced-motion support.
GitHub protection	Automated CI on pushes/PRs, ignored private state, backup-on-upgrade, health endpoint and no automatic deployment job.

Scope preserved: event CRUD, seat sections and blocks, student registration, booking history and cancellation, event/booking PDF and CSV exports, notification emails and free RMB/CNY admission. No payment processing.

Architecture, integrity and migration

The migration respects the original V5.2 dataset rather than replacing it.

Database changes

Users: `password_ready` and `auth_version` support activation and password-session revocation.

Events: `booking_enabled` and `checkin_enabled` provide separate admin switches.

Tickets: new Ticket entity links exactly one seat Booking to one registered attendee, includes a stable random nonce, entry status, checked-in time and operator ID.

Integrity: partial unique indexes protect active seats and active/checked-in tickets per student/event; cancellation triggers revoke QR tickets.

V5.2 compatibility and backup

Before changing an old SQLite schema, the application uses the SQLite backup API to create a checked backup under private instance/backups/. The migration is idempotent: reruns do not rotate QR nonces or delete any booking rows. Existing admins must activate a password once.

When a V5.2 student owns several active seats in one event, only the first can be assigned to that same student. Extra seats remain reserved but their QR tickets are marked **unassigned**, pending admin verification of another registered student. This prevents mass issuance of duplicate student admission passes.

Gate and QR security

• QR contains a signed ticket ID and nonce, **not** a name, email or Student ID.

• GET verification is read-only; staff/admin must perform CSRF-protected POST to check in.

• Conditional database UPDATE rejects duplicate check-ins, including competing gate scanners.

• Production must preserve a persistent independent TICKET_SIGNING_KEY. Admin password activation changes invalidate older admin sessions.

• Encrypted transit (HTTPS), staff identity checks and institutional privacy controls remain operational prerequisites.

Setup, validation and release

Windows PowerShell quick start

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python -m flask --app run.py init-db
python -m flask --app run.py seed-events
python run.py
```

First admin: copy the generated 20-minute key from PowerShell into **/admin/activate**, choose a password and log in thereafter at **/admin/login**. No application file or Notepad key editing. Expired activation key: `flask --app run.py admin-key EMAIL`. Already activated password recovery: `flask --app run.py admin-reset EMAIL`.

Checks and remaining validation

Offline checks: Python compilation, 27 Jinja template parsings, JavaScript syntax, SVG validation and an execution of the actual SQLite V5.3 migration (twice) including ticket uniqueness, legacy backfill and cancellation triggers. A complete Flask integration test suite is included but cannot be executed in this runner because Flask dependencies are unavailable and package downloads are blocked. Run the suite before deployment.

```
python -m unittest discover -s tests -v
```

GitHub deployment safety

The repository contains a read-only GitHub Actions CI workflow; it does not deploy. A push to GitHub alone does not mutate the running SQLite file. Require CI, review and a separate backed-up migration/deployment step before accepting a release. Keep .env, private keys, databases, backups and exported student reports outside version control.

Release information

GitHub tag	v5.3
Release title	CSTBS v5.3 - Smart University Stadium Ticketing
Main documentation	README, CHANGELOG, SECURITY, user/admin/deployment manuals, UML, database design, GitHub release notes